LAPORAN PRAKTIKUM

POSTTEST 1

ALGORITMA PEMROGRAMAN LANJUT

Disusun oleh:

Setriyani (2509106039)

Kelas (A2 ‘25)

PROGRAM STUDI INFORMATIKA

UNIVERSITAS MULAWARMAN

SAMARINDA

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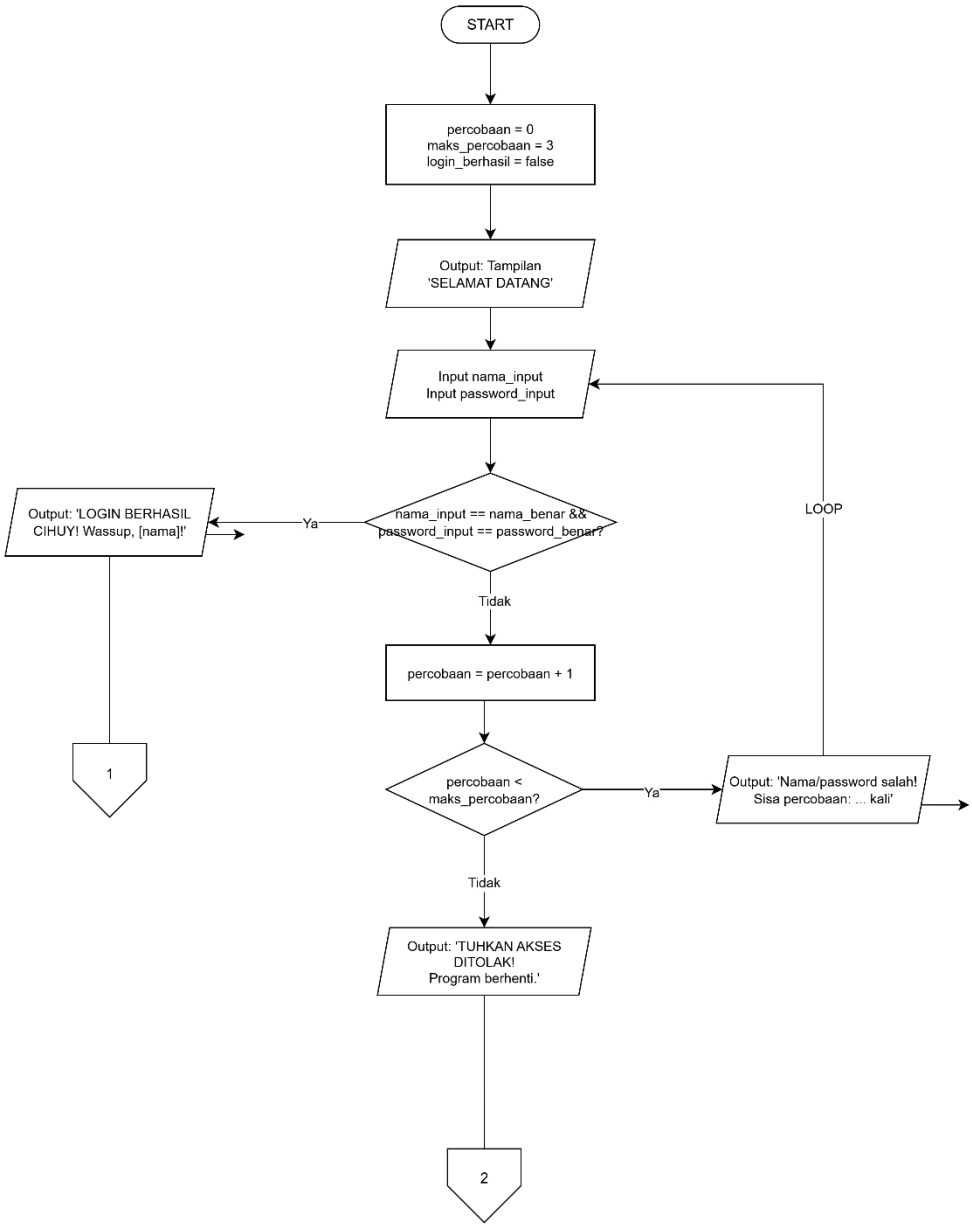
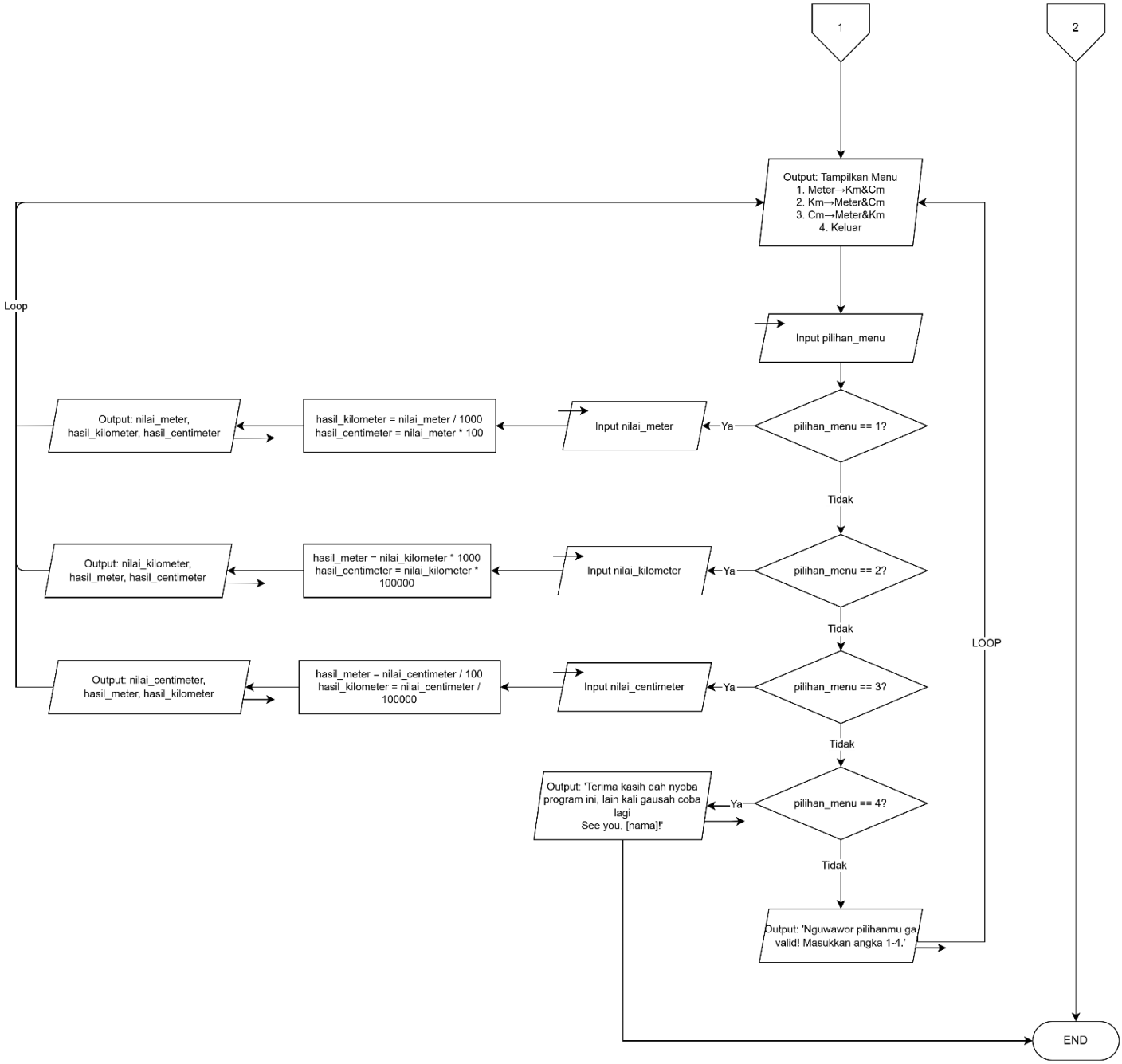
1. Flowchart



Gambar 1.1 Bagian login

Catatan sedikit:

Maaf ya abang/mba untuk bagian loginnya ga konsisten gitu arah panahnya, dikarenakan kalau saya ngikutin arah konsistennya dengan naruh “ya” di arah kiri flowchartnya jadi kurang rapi karena loopnya bakal nabrak



Gambar 1.2 Bagian Menu

Penjelasan Flowchart:

Program mulai, langsung set variabel awal. Terus tampil tulisan selamat datang, lalu minta user input nama dan password. Setelah diinput, dicek dulu bener ga nama dan passwordnya. Kalau bener, langsung output login berhasil. Kalau salah, percobaan ditambah 1, terus dicek udah 3x belum. Kalau belum 3x, output pesan salah, balik lagi ke input nama password. Kalau udah 3x, output akses ditolak, program selesai. Terus lanjut ke menu pilihan, user input angka 1-4. Dicek satu-satu dari atas ke bawah:

- Pilih 1 - input meter - hitung - output hasil
- Pilih 2 - input kilometer - hitung - output hasil
- Pilih 3 - input centimeter - hitung - output hasil
- Pilih 4 - output terima kasih - END
- Selain itu - output pilihan ga valid

2. Deskripsi Singkat Program

Program ini pertama-tama nyimpen dulu nama dan password yang bener di variabel, yaitu "Setri" dan "039". Terus program nampilin layar selamat datang, lalu minta user masukin nama dan password. Setiap kali salah, percobaan ditambah 1 dan muncul pesan sisa percobaan. Kalau udah salah 3 kali, program langsung berhenti. Kalau benar, lanjut ke menu.

Di menu, user disuruh pilih angka 1 sampai 4. Program ngecek pilihan satu-satu pakai if-else. Kalau pilih 1, user masukin nilai meter, terus program ngitung hasil kilometernya dengan dibagi 1000 dan centimeternya dengan dikali 100, lalu ditampilin hasilnya. Kalau pilih 2, user masukin kilometer, dikaliin 1000 buat dapet meter dan dikaliin 100000 buat dapet centimeter. Kalau pilih 3, user masukin centimeter, dibagi 100 buat meter dan dibagi 100000 buat kilometer. Kalau pilih 4, muncul pesan keluar dan program selesai. Kalau inputnya angka lain, muncul pesan pilihan tidak valid. Semua ini diulang terus sampai user milih 4.

3. Source Code

```
#include <iostream>
#include <string>
#include <iomanip>
using namespace std;

int main() {

    string nama_benar    = "Setri";
    string password_benar = "039";

    string nama_input;
    string password_input;

    int percobaan      = 0;
    int maks_percobaan = 3;
    bool login_berhasil = false;

    cout << "========================================" << endl;
    cout << "         SELAMAT DATANG                 " << endl;
    cout << "     PROGRAM KONVERSI PANJANG           " << endl;
    cout << "========================================" << endl;
    cout << endl;

    while (percobaan < maks_percobaan) {
        percobaan++;

        cout << "Percobaan ke-" << percobaan << " dari " << maks_percobaan
<< endl;
        cout << "----------------------------------------" << endl;
        cout << "Nama     : ";
        cin  >> nama_input;
        cout << "Password : ";
        cin  >> password_input;
```

```
        if (nama_input == nama_benar && password_input == password_benar)
{
            login_berhasil = true;
            break;
        } else {
            if (percobaan < maks_percobaan) {
                cout << endl;
                cout << "  Nama atau passwordmu salah kocak!" << endl;
                cout << "      Sisa percobaan: " << (maks_percobaan -
percobaan) << " kali" << endl;
                cout << endl;
            }
        }
    }

    if (!login_berhasil) {
        cout << endl;
        cout << "=========================================" << endl;
        cout << "  TUHKAN AKSES DITOLAK!                  " << endl;
        cout << " Kamu udah 3x salah, Programnya berhenti deh " << endl;
        cout << "=========================================" << endl;
        return 0;
    }

    cout << endl;
    cout << "======================================" << endl;
    cout << "   LOGIN BERHASIL CIHUY! Wassup, " << nama_benar << "!    "
<< endl;
    cout << "======================================" << endl;
    cout << endl;

    int pilihan_menu = 0;

    while (pilihan_menu != 4) {

        cout << "========================================" << endl;
        cout << "         MENU KONVERSI PANJANG          " << endl;
        cout << "========================================" << endl;
        cout << "  1. Meter      --> Kilometer & Centimeter" << endl;
        cout << "  2. Kilometer  --> Meter    & Centimeter" << endl;
        cout << "  3. Centimeter --> Meter    & Kilometer " << endl;
        cout << "  4. Keluar                              " << endl;
        cout << "========================================" << endl;
        cout << "  Pilih cepat. : ";
        cin  >> pilihan_menu;
        cout << endl;

        if (pilihan_menu == 1) {

            double nilai_meter = 0;
            cout << "  Masukin nilai dalam Meter: ";
            cin  >> nilai_meter;

            double hasil_kilometer  = nilai_meter / 1000.0;
```

```
        double hasil_centimeter = nilai_meter * 100.0;

        cout << endl;
        cout << "  ---- HASIL KONVERSI ----" << endl;
        cout << fixed << setprecision(4);
        cout << "  " << nilai_meter   << " Meter" << endl;
        cout << "    = " << hasil_kilometer  << " Kilometer" << endl;
        cout << "    = " << hasil_centimeter << " Centimeter" << endl;
        cout << "  ------------------------" << endl;
        cout << endl;

    } else if (pilihan_menu == 2) {

        double nilai_kilometer = 0;
        cout << "  Masukin nilai dalam Kilometer: ";
        cin  >> nilai_kilometer;

        double hasil_meter      = nilai_kilometer * 1000.0;
        double hasil_centimeter = nilai_kilometer * 100000.0;

        cout << endl;
        cout << "  ---- HASIL KONVERSI ----" << endl;
        cout << fixed << setprecision(4);
        cout << "  " << nilai_kilometer  << " Kilometer" << endl;
        cout << "    = " << hasil_meter      << " Meter" << endl;
        cout << "    = " << hasil_centimeter << " Centimeter" << endl;
        cout << "  ------------------------" << endl;
        cout << endl;

    } else if (pilihan_menu == 3) {

        double nilai_centimeter = 0;
        cout << "  Masukin nilai dalam Centimeter: ";
        cin  >> nilai_centimeter;

        double hasil_meter     = nilai_centimeter / 100.0;
        double hasil_kilometer = nilai_centimeter / 100000.0;

        cout << endl;
        cout << "  ---- HASIL KONVERSI ----" << endl;
        cout << fixed << setprecision(6);
        cout << "  " << nilai_centimeter << " Centimeter" << endl;
        cout << "    = " << hasil_meter      << " Meter" << endl;
        cout << "    = " << hasil_kilometer  << " Kilometer" << endl;
        cout << "  ------------------------" << endl;
        cout << endl;

    } else if (pilihan_menu == 4) {

        cout << "========================================" << endl;
        cout << "  Terima kasih dah nyoba program ini      " << endl;
        cout << "  lain kali gausah nyoba lagi             " << endl;
        cout << "  see you  " << nama_benar << "!          " << endl;
        cout << "========================================" << endl;
        cout << endl;
```

```
        } else {

            cout << "  Nguwawor Pilihanmu ga valid! Masukkan angka 1-4."
<< endl;
            cout << endl;

        }
    }

    return 0;
}
```

4. Hasil Output

```
=========================================
         SELAMAT DATANG
     PROGRAM KONVERSI PANJANG
=========================================

Percobaan ke-1 dari 3
-----------------------------------------
Nama     :
```

Gambar 4.1
Tampilan login

```
=========================================
         SELAMAT DATANG
     PROGRAM KONVERSI PANJANG
=========================================

Percobaan ke-1 dari 3
-----------------------------------------
Nama     : budi
Password : 999

  Nama atau passwordmu salah kocak!
      Sisa percobaan: 2 kali
```

Gambar 4.2 Apabila
Nama/Pw salah

```
Percobaan ke-3 dari 3
-----------------------------------------
Nama     : y
Password : y

=========================================
  TUHKAN AKSES DITOLAK!
 Kamu udah 3x salah, Programnya berhenti deh
=========================================
```

Gambar 4.3 sudah
3X salah

```
=========================================
          SELAMAT DATANG
      PROGRAM KONVERSI PANJANG
=========================================

Percobaan ke-1 dari 3
-----------------------------------------
Nama     : Setri
Password : 039

=========================================
   LOGIN BERHASIL CIHUY! Wassup, Setri!
=========================================
```

Gambar 4.4 Login
Berhasil

```
=========================================
        MENU KONVERSI PANJANG
=========================================
  1. Meter      --> Kilometer & Centimeter
  2. Kilometer  --> Meter     & Centimeter
  3. Centimeter --> Meter     & Kilometer
  4. Keluar
=========================================
  Pilih cepat. :
```

Gambar 4.5
Tampilan Menu

```
=========================================
        MENU KONVERSI PANJANG
=========================================
  1. Meter      --> Kilometer & Centimeter
  2. Kilometer  --> Meter     & Centimeter
  3. Centimeter --> Meter     & Kilometer
  4. Keluar
=========================================
  Pilih cepat. : 1

  Masukin nilai dalam Meter: 4

  ---- HASIL KONVERSI ----
  4.0000 Meter
    = 0.0040 Kilometer
    = 400.0000 Centimeter
  ------------------------
```

Gambar 4.6 Pilihan 1

```
========================================
        MENU KONVERSI PANJANG
========================================
  1. Meter      --> Kilometer & Centimeter
  2. Kilometer  --> Meter     & Centimeter
  3. Centimeter --> Meter     & Kilometer
  4. Keluar
========================================
  Pilih cepat. : 2

  Masukin nilai dalam Kilometer: 4

  ---- HASIL KONVERSI ----
  4.0000 Kilometer
    = 4000.0000 Meter
    = 400000.0000 Centimeter
  -----------------------
```

Gambar 4.7 Pilihan 2

```
========================================
        MENU KONVERSI PANJANG
========================================
  1. Meter      --> Kilometer & Centimeter
  2. Kilometer  --> Meter     & Centimeter
  3. Centimeter --> Meter     & Kilometer
  4. Keluar
========================================
  Pilih cepat. : 3

  Masukin nilai dalam Centimeter: 4

  ---- HASIL KONVERSI ----
  4.000000 Centimeter
    = 0.040000 Meter
    = 0.000040 Kilometer
  -----------------------
```

Gambar 4.8 Pilihan 3

```
=======================================
         MENU KONVERSI PANJANG
=======================================
  1. Meter      --> Kilometer & Centimeter
  2. Kilometer  --> Meter    & Centimeter
  3. Centimeter --> Meter    & Kilometer
  4. Keluar
=======================================
  Pilih cepat. : 4

=======================================
  Terima kasih dah nyoba program ini
  lain kali gausah nyoba lagi
  see you  Setri!
=======================================
```

Gambar 4.9 Pilihan 4

```
=======================================
         MENU KONVERSI PANJANG
=======================================
  1. Meter      --> Kilometer & Centimeter
  2. Kilometer  --> Meter    & Centimeter
  3. Centimeter --> Meter    & Kilometer
  4. Keluar
=======================================
  Pilih cepat. : 5

  Nguwawor Pilihanmu ga valid! Masukkan angka 1-4.
```

Gambar 4.10
Pilihan diluar dari
menu

5. Langkah-langkah GIT

5.1 GIT Init

```
PS C:\Users\ACER\OneDrive\Documents\Praktikum-apl> git init
Reinitialized existing Git repository in C:/Users/ACER/OneDrive/Documents/Praktikum-apl/.git/
```

Perintah buat mulai nge-track project pake Git. Jalanin sekali aja di awal.

5.1 GIT Add

```
PS C:\Users\ACER\OneDrive\Documents\Praktikum-apl> git add .
```

Milih file mana yang mau disimpan di tahap berikutnya.

5.2 GIT Commit

```
PS C:\Users\ACER\OneDrive\Documents\Praktikum-apl> git commit -m "first commit"
[main (root-commit) e687ac1] first commit
 4 files changed, 202 insertions(+)
 create mode 100644 Kelas/Pertemuan 1/Pertemuan1.cpp
 create mode 100644 Kelas/Pertemuan 1/Pertemuan1.exe
 create mode 100644 Post-test/Post-test-1/2509106039-Setriyani-PT1.cpp
 create mode 100644 Post-test/Post-test-1/2509106039-Setriyani-PT1.exe
```

Nyimpen perubahan yang udah di add tadi, lengkap sama pesannya.

5.1 GIT Remote

```
PS C:\Users\ACER\OneDrive\Documents\Praktikum-apl> git remote add origin https://github.com/setryyni/Praktikum-Apl.git
```

Nyambungin project lokal ke repository GitHub.

5.3 GIT Push

```
PS C:\Users\ACER\OneDrive\Documents\Praktikum-apl> git push -u origin main
Enumerating objects: 10, done.
Counting objects: 100% (10/10), done.
Delta compression using up to 4 threads
Compressing objects: 100% (8/8), done.
Writing objects: 100% (10/10), 933.59 KiB | 1.84 MiB/s, done.
Total 10 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), done.
To https://github.com/setryyni/Praktikum-Apl.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
```

Upload semua commit ke repository.